Select digitisation exposure with Camera-41_exp

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Work instruction for selecting digital-camera shutter time when photographing C-41 negatives, at a fixed chosen aperture, using film base plus fog RAWs.

1 Applications used

SpectraLab measures the illumination emission spectrum and the colour chart patch reflectance spectra, using an instrument such as an X-Rite i1Pro 2. Measurements are exported as spectral data to Camera-41_DC and Camera-41_pre.
https://github.com/chto0703009/SpectraLab

Camera-41_exp helps select the digital-camera exposure for digitising C-41 negatives, using RAW images of film base plus fog, FB+F. It exports no JSON handoff; new images are captured at the selected setting.
https://github.com/chto0703009/Camera-41_exp

Camera-41_DC characterises the digital camera using original RAW images of ColorChecker Digital SG 140, measured reflectance spectra of the chart and the capture-light emission spectrum. The result is a reviewed camera profile that can be imported into Camera-41_pre.
https://github.com/chto0703009/Camera-41_DC

Camera-41_pre characterises C-41 film using grey and colour-chart exposure series and their associated measurements. It uses the camera profile from Camera-41_DC and prepares the film characterisation for Camera-41_neg.
https://github.com/chto0703009/camera-41_pre

Camera-41_neg uses the film characterisation from Camera-41_pre to invert digitised C-41 negatives into positive images.
https://github.com/chto0703009/camera-41_neg

Capture a RAW series of the same unexposed, processed film area, FB+F, at different shutter times. Camera-41_exp compares signals and headroom in the four Bayer channels and proposes a tested time. Set the camera yourself and confirm it with a new capture. The app is standalone and requires no film or camera profile. For subsequent PRE work, deliver new FB+F and dark images directly; no JSON is transferred from EXP.

The Camera-41_exp main window

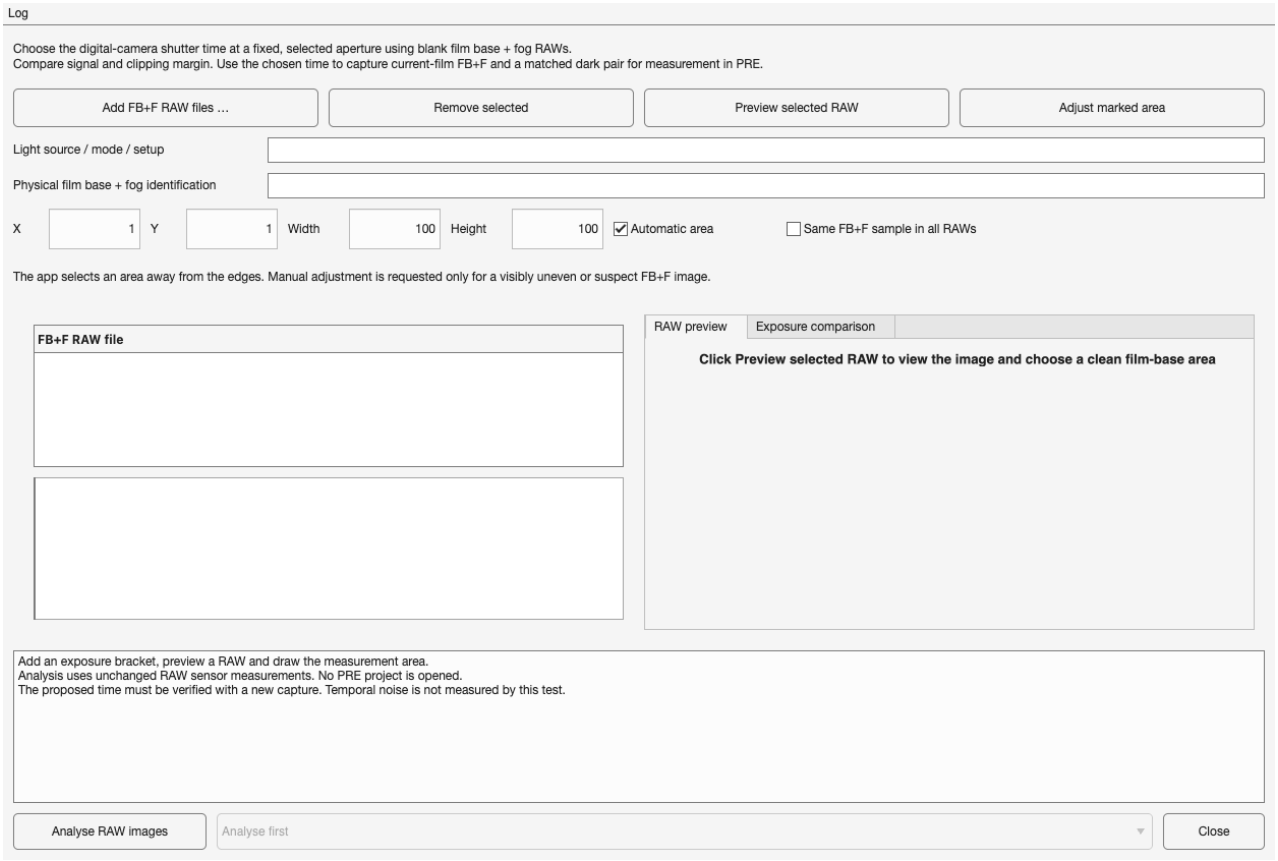


Figure 1. Camera-41_exp v0.1.4 with no RAW files loaded. Captured from the locally installed application on 1 November 2026.

At the top, add FB+F RAWs and describe lighting setup and physical film sample. Automatic area uses a region away from image edges. Files and results appear on the left. On the right, switch between RAW preview and Exposure comparison. Analyse RAW images and the passing-time selection are at the bottom.

Workflow

Select film sample, ISO and aperture → capture a shutter-time RAW series → import and identify the sample → review measurement area → analyse signal and headroom → add captures as needed → select a tested time → capture new FB+F and two matched darks → measure in Camera-41_pre.

What the selection applies to

Exposure here means the digital camera photographing processed film. It does not change the original analogue film exposure. The app measures neither temporal dark noise nor complete S/N and provides no production approval. A bright preview does not establish usable RAW signal.

2 Prepare and capture the FB+F series

Choose the correct physical film sample

Use a clean, unexposed, processed part of the film to digitise. It must contain FB+F without scene content, text, perforations, scratches, dust or holder edges in the measurement area. A light-exposed, dense leader is unsuitable. Every comparison image must show the same physical film area.

FB+F normally transmits more light than denser scene regions and is therefore an important reference for avoiding excessive digitisation exposure. Subsequent work must also assess S/N in dense scene areas. FB+F alone does not prove adequate S/N for every scene.

Lock the setup and choose aperture

Use the intended camera, lens, holder and digitisation light. Stabilise illumination, exclude stray light, align the camera perpendicular to the film plane, and lock focus and geometry. Select aperture and ISO first. Keep body, ISO, aperture, light mode, sample and framing fixed throughout. Vary only shutter time.

Record roll/sample ID, camera identity, lens, ISO, aperture, light unit, actual mode and date. EXP needs no spectral files for this comparison. Later light compatibility with the film characterisation is checked in PRE or NEG using their measured-light inputs.

Capture several shutter-time settings

Start conservatively and take progressively longer exposures, preferably one-third stops near the limit. An adequately bracketed proposal needs passing times and a higher nonpassing capture. Internal proportionality checking needs at least three distinct passing times spanning a factor of two. Add captures as directed by the result; one image does not establish the entire comparison.

The digital camera's actual shutter time was not verified by separate timing measurements in this work. Because the camera controls timing electronically, its selected time recorded in RAW metadata was used as the timing input. This is a methodological choice, not a measured shutter calibration. The selected digitisation exposure must still be checked for RAW-signal proportionality, clipping headroom and applicable noise evidence. EXP compares RAW signals at the recorded times; noise and S/N are measured subsequently in PRE.

Keep camera-original RAW files

Preserve original RAWs unchanged and record sample and file identities. JPEG or rendered TIFF may have changed white balance, tone curves, channel values and clipping and cannot replace the input. EXP uses Bayer measurements and metadata for R, G1, G2 and B, not screen brightness, for acceptance.

3 Import the series and review the area

Start and add RAWs

Open the complete Camera-41_exp folder in MATLAB. Open start.m and press Run, or select the app as Current Folder and run startCamera41_exp. Let local Python and RAW-reader checks finish. PRE and NEG need not be installed or open. Manual region editing uses Image Processing Toolbox in the documented installation.

Choose Add FB+F RAW files … and select the series originals. Further choices are added without duplicating the same path. Remove selected removes a row from the series, not the original file. The chooser remembers its last RAW folder after restart. Ensure each file remains accessible on its drive.

Describe the light and sample

App field	What to enter
Light source / mode / setup	Identify actual light unit, mode and setup sufficiently to distinguish the sample from other series.
Physical film base + fog identification	Film type, roll/sample ID and the physical FB+F area photographed.
Same FB+F sample in all RAWs	Confirm only after checking that every file shows the same physical sample with unchanged setup.

The app checks matching camera body, ISO, aperture and image dimensions across the series. Metadata cannot verify the film sample or light mode; check and document these yourself.

Use a representative measurement area

Select a file and choose Preview selected RAW. Automatic area normally uses the central 70% of width and height, with approximately 15% margins on each side. The same region is used for the series. RAW preview is brightness-adjusted for viewing, not a measurement result.

Suspect structure, strong light variation or low signal triggers a warning and yellow region. Inspect the film. Use Adjust marked area to move or resize it to clean, representative FB+F. X, Y, Width and Height refer to original visible-sensor pixels. Do not exclude a bright clipped part of actual FB+F merely to obtain PASS. Recapture if the correct sample cannot be measured.

Run Analyse RAW images. Repeat after changing files, sample details or area. Automatic uniformity checking supports review but cannot reliably identify physical film content.

4 Assess results and choose a time

Review all four Bayer channels

Exposure comparison plots RAW signal and clipping headroom against time for R, G1, G2 and B. The table shows file, time, status, minimum headroom and greatest channel fraction at/above the limit. Read the channel explanation. A correctly decoded RAW can give FAIL; this does not mean file import failed.

Status	Interpretation and action
PASS	All four channels have at least 1/3 EV headroom at the 99.9th percentile, and no channel reaches the FAIL threshold for fraction at/above the linearity limit. The time may enter selection, but the overall series still needs review.
REVIEW	No channel triggers FAIL, but at least one has less than 1/3 EV headroom. Do not increase exposure to improve the preview; try a shorter time.
FAIL	At least one channel has 0.01% or more values at/above the linearity limit. First check sample and area. If correct, use a shorter time.

The limit applies to RAW after black subtraction using the RAW reader's reported linearity limit. It is neither a JPEG histogram nor a separate physical sensor-linearity measurement. Positive EV headroom means the percentile lies below the limit.

Complete the series for a usable selection

The app proposes the longest tested passing time. Without a higher nonpassing capture, the proposal is provisional: take a longer trial. If the next higher nonpassing time is more than approximately 1/3 EV away, refine the bracket. Use a time actually captured and accepted, not a calculated but untested setting.

Review proportionality separately. It compares signal divided by time for passing captures, using an internal 5% tolerance from the channel median in this version. At least three distinct passing times spanning a factor of two are required. Insufficient evidence means more captures are needed. Investigate nonproportional-response warnings before using the candidate; one PASS row does not resolve them.

Finally select a tested passing time in the bottom list and record time, source, sample ID, ISO, aperture, light and assessment. Investigate inconsistent status in repeats at the same time. A recommended time is an exposure choice, not full camera/film qualification.

5 Confirm the choice and deliver to Camera-41_pre

Take new captures at the selected setting

Set the camera yourself to the selected time. Preserve camera, ISO, aperture, light, RAW format, focus and geometry. Capture a new FB+F image from the current film as confirmation. Also take two independent digital dark RAWs with matching settings and light blocked from the sensor. Match relevant conditions; do not duplicate one dark file.

An unexposed, processed C-41 frame is FB+F, not a digital dark. The two darks are needed for subsequent noise measurement in PRE; they are not imported as a separate dark series into the EXP main window.

What to hand over

New input	Delivery to the next step
FB+F confirmation capture	Original RAW from the current film at the chosen time. Identify physical roll and sample area.
Two dark captures	Two independent original RAWs matching body, ISO, time, aperture and other relevant conditions.
Trial record	Selected time, camera/lens, ISO, aperture, light unit/mode, date, operator, sample ID and filenames. Retain the comparison series for traceability.

Import the new FB+F and dark captures directly into PRE tab 7 for digital-camera exposure. Check a shared sensor area and measure noise and S/N as described in the PRE instruction. Results accompany the film characterisation to NEG. EXP has no JSON export for this handoff and need not be open when the next app is used.

What Camera-41_exp saves

Log → Open complete log opens logs/camera41_exp.jsonl. The local log contains file selections, setup details, analyses, time choices, durations and errors. Original RAWs are unchanged. The last RAW folder is remembered, but the file list and other window settings are not restored after closing. Keep the original series and your own record; the log is not a portable PRE delivery.

Reassess exposure when light, film base, camera, ISO, aperture or geometry changes. Routine import of a new roll then uses its applicable film/camera configuration and own FB+F, as described in the NEG instruction.

Handbook and references

Checked against Camera-41_exp v0.1.4 and the Swedish handbook text on 1 November 2026. Use the section titles below if pagination changes in a later edition.

Open the English handbook PDF on GitHub

Open the English handbook source on GitHub

For the introduction and section 2, see the handbook sections on why to use EXP, and capturing images and selecting the measurement area. They explain FB+F and keeping the sample/setup fixed.

For instruction section 3, see installation/startup and image capture/measurement-area selection, particularly import/preview, automatic area and warnings requiring adjustment.

For instruction section 4, see the handbook's results discussion: PASS, REVIEW and FAIL; plots and table; provisional proposals; and choosing a time with margin.

For instruction section 5, see the handbook's post-trial workflow, logging and help, particularly continuing in PRE with current captures and saved information. The English PDF is in output/pdf/Camera41_exp_Handbook_EN.pdf in the app folder.

For subsequent measurements, follow the separate PRE instruction and handbook chapters 7 on digital-camera exposure and 18 on S/N with matched darks. The NEG instruction covers routine roll import.

Inputs, thresholds and time selection were also checked against current import, area and selection functions. RAW-reader limits, automatic uniformity screening and proportionality checks support this task; they are not new physical calibration or production qualification.

Appendix A Equipment used in the documented work

The table describes the retained FB+F/digitisation setup underlying the exposure method. Distinguish it from a new EXP trial, which needs its own record of actual film sample, light mode and settings. The image in this instruction shows startup, not a new exposure result.

Step	Documented equipment and use
Exposure-test film sample	The retained Portra 160 evidence describes clean, unexposed, processed FB+F, captured at the same film plane and framing at several times. It is not a chart test.
Camera	Existing digitisation camera: Canon EOS 5DS R body 063021000065, identified in saved camera and film deliveries.
Lens	Canon EF100mm f/2.8 Macro USM according to the saved Canon project. Its camera identity gives no lens serial number.
Holder and digitisation light	VALOI setup and CS Lite; the retained Portra series uses WHITE. CS Lite B is identified as digitisation light in the documented setup. Confirm exact holder model from the original record.
Fixed settings and tested times	The retained Portra series uses ISO 100 and f/8. Tests at 1/13 and 1/10 s gave PASS, 1/8 s REVIEW, and 1/6–1/5 s FAIL. The operator chose 1/10 s on 14 August 2026. These are historical test results, not general times for other films or lights.
Digital darks after time selection	Retained evidence identifies two separate dark RAWs at ISO 100, f/8 and 1/8 s. They belong to that setting and cannot replace new darks at, for example, 1/10 s. Use the correctly matched pair in the next measurement step.
Software for this instruction	Camera-41_exp v0.1.4 in MATLAB R2025b with bundled LibRaw reader. The app imports the series, supports area review and selects a tested time; camera settings are applied manually.

Equipment and trial details were checked against saved Canon camera identity, film-delivery configurations and FILM_BASE_RAW_CAPTURE.md. No specific support or light-tight dark-cover manufacturer is securely documented there. No new equipment identities or test results have been assumed.